

Statistical Optimization: Lecture 8

Line Search Methods: Gradient Descent Step Size Selection

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Step size in gradient descent

For gradient descent, the update takes the form

$$\theta_{t+1} = \theta_t - \gamma_t \nabla f(\theta_t).$$

In theoretical analysis, one often chooses a fixed step size, for example

$$\gamma_t \equiv \frac{1}{L},$$

when f is L -smooth.

This choice is simple, guarantees descent, and is convenient for proving convergence.

Objective

In practice, however, a fixed step size may not be the best choice.

If the step size is too small:

- the method is stable,
- but progress can be very slow.

If the step size is too large:

- the iterates may oscillate,
- or even fail to decrease the objective.

So it is natural to ask: *how should we choose the step size at each iteration?*

Outline

Line search

Wolfe conditions

Backtracking method

Line search setup

To choose the step size adaptively, we consider the more general update

$$\theta_{t+1} = \theta_t + \gamma_t p_t,$$

where p_t is a search direction and $\gamma_t > 0$ is the step size.

For gradient descent, we take

$$p_t = -\nabla f(\theta_t).$$

Then define the one-dimensional function

$$\phi(\gamma_t) := f(\theta_t + \gamma_t p_t), \quad \gamma_t > 0.$$

Thus, each iteration reduces the multidimensional problem to a one-dimensional search along the direction p_t .

Exact line search

Once the direction p_t is chosen, the step length γ_t can be obtained by considering

$$\min_{\gamma_t > 0} \phi(\gamma_t) = \min_{\gamma_t > 0} f(\theta_t + \gamma_t p_t).$$

If we solve this one-dimensional problem exactly, we derive the maximum benefit from the direction p_t .

In practice, this may be done by bisection search, golden-section search and other methods.

Exact line search

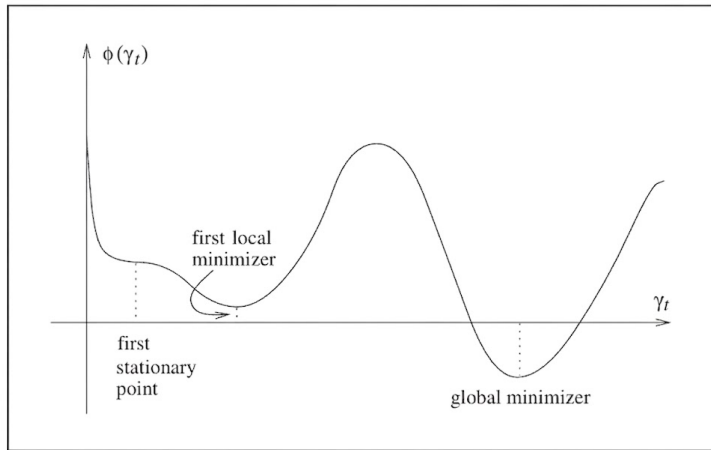


Figure: The ideal step length is the global minimizer.

Inexact line search

However, exact minimization may be expensive and is usually unnecessary.

In practice, we only seek a step length that gives a sufficient decrease of the objective and has reasonable practical cost.

So the basic question in line search is:

How can we define conditions that make a step length acceptable?

This leads to the termination conditions for line search, starting with the Wolfe conditions.

Outline

Line search

Wolfe conditions

Backtracking method

Sufficient decrease condition

A popular inexact line search condition first requires **sufficient decrease**:

$$f(\theta_t + \gamma_t p_t) \leq f(\theta_t) + c_1 \gamma_t \nabla f(\theta_t)^\top p_t,$$

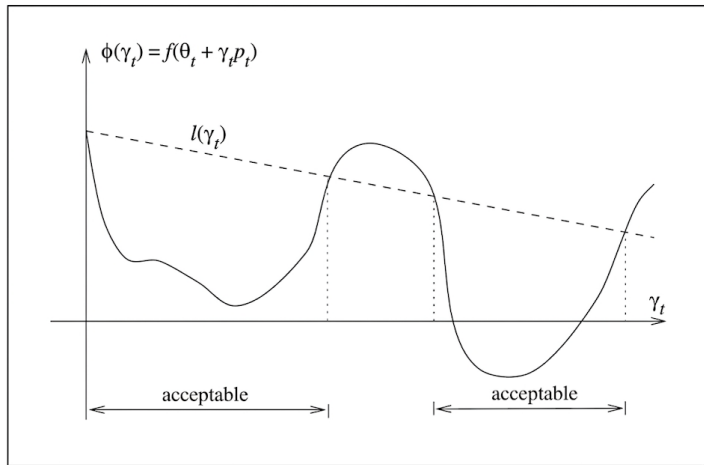
for some constant $c_1 \in (0, 1)$.

This inequality is sometimes called **Armijo condition**.

Remark.

- The reduction in f should be proportional to both the step length γ_t and the directional derivative $\nabla f(\theta_t)^\top p_t$.
- The right-hand side is a linear function of γ_t ; denote it by $l(\gamma_t)$.
- This condition alone is not enough, since it is satisfied for all sufficiently small γ_t .

Sufficient decrease condition



Curvature condition

To rule out unacceptably short steps, we introduce a second requirement, called the **curvature condition**:

$$\nabla f(\theta_t + \gamma_t \mathbf{p}_t)^\top \mathbf{p}_t \geq c_2 \nabla f(\theta_t)^\top \mathbf{p}_t,$$

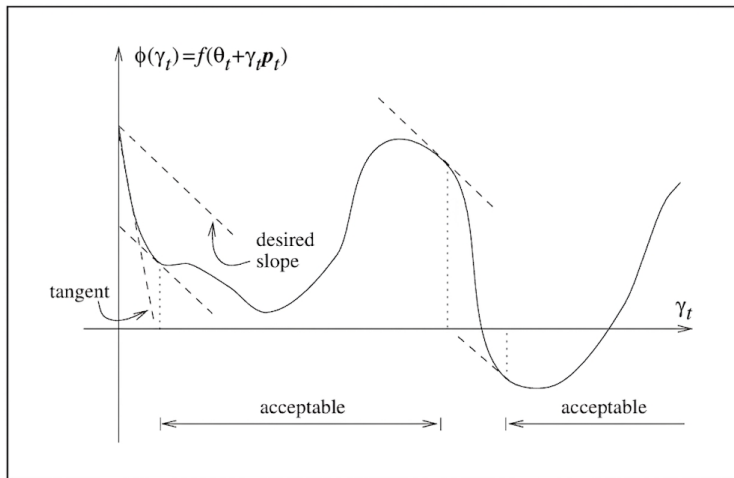
for some constant $c_2 \in (c_1, 1)$.

Since

$$\phi'(\gamma_t) = \nabla f(\theta_t + \gamma_t \mathbf{p}_t)^\top \mathbf{p}_t,$$

the curvature condition ensures that the slope of ϕ at γ_t is greater than c_2 times the initial slope $\phi'(0)$.

Curvature condition



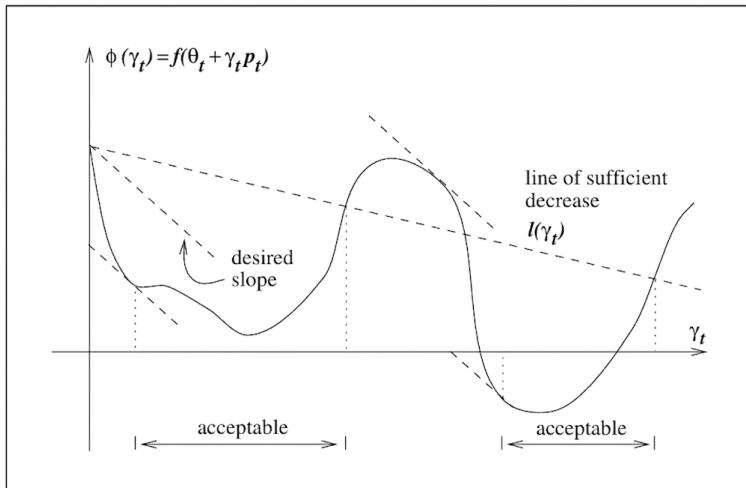
Wolfe conditions

Wolfe conditions. Together, the sufficient decrease and curvature conditions are called the **Wolfe conditions**:

$$\begin{cases} f(\theta_t + \gamma_t \mathbf{p}_t) \leq f(\theta_t) + c_1 \gamma_t \nabla f(\theta_t)^\top \mathbf{p}_t, \\ \nabla f(\theta_t + \gamma_t \mathbf{p}_t)^\top \mathbf{p}_t \geq c_2 \nabla f(\theta_t)^\top \mathbf{p}_t, \end{cases} \quad 0 < c_1 < c_2 < 1.$$

The first condition enforces sufficient decrease, while the second prevents the step length from being too short.

Wolfe conditions



Strong Wolfe conditions

A commonly used variant is the **strong Wolfe conditions**:

$$\begin{cases} f(\theta_t + \gamma_t \mathbf{p}_t) \leq f(\theta_t) + c_1 \gamma_t \nabla f(\theta_t)^\top \mathbf{p}_t, \\ |\nabla f(\theta_t + \gamma_t \mathbf{p}_t)^\top \mathbf{p}_t| \leq c_2 |\nabla f(\theta_t)^\top \mathbf{p}_t|, \end{cases} \quad 0 < c_1 < c_2 < 1.$$

The strong Wolfe conditions require not only that the slope should not remain too negative, but also that even if it becomes positive, it should not be too large.

Equivalently, the directional derivative at the new point must be small in magnitude.

Outline

Line search

Wolfe conditions

Backtracking method

Sufficient decrease and backtracking

We have mentioned that the sufficient decrease condition alone is not enough to ensure that the algorithm makes reasonable progress along the search direction. Indeed, for all sufficiently small $\gamma_t > 0$, the condition

$$f(\theta_t + \gamma_t p_t) \leq f(\theta_t) + c_1 \gamma_t \nabla f(\theta_t)^\top p_t$$

is automatically satisfied.

However, if the candidate step lengths are chosen appropriately, we can use this condition alone to terminate the line search.

This leads to the **backtracking** strategy.

Backtracking line search

Algorithm

Input: current iterate θ_t , search direction p_t , initial trial step length $\bar{\gamma} > 0$

Parameters: $\rho \in (0, 1)$, $c_1 \in (0, 1)$

1. Set $\gamma \leftarrow \bar{\gamma}$.

2. While

$$f(\theta_t + \gamma p_t) > f(\theta_t) + c_1 \gamma \nabla f(\theta_t)^\top p_t,$$

set

$$\gamma \leftarrow \rho \gamma.$$

3. Return $\gamma_t = \gamma$.

Backtracking line search

Remark.

1. Starting from an initial trial step length, we repeatedly shorten it until the sufficient decrease condition is satisfied.
2. Similar ideas can also be used in line search procedures based on the Wolfe conditions.

Summary

Method		Criterion	Strength	Weakness
Exact search	line	Minimize the objective exactly along the line.	Best possible decrease for the given direction.	High computational cost.
Wolfe	Condi- tions	Sufficient decrease + curvature condition.	Good practical and theoretical balance.	(Most used)